

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 5-01-111

WASTE DISCHARGE REQUIREMENTS
FOR
LAKE COUNTY SANITATION DISTRICT
NORTHWEST REGIONAL WASTEWATER SYSTEM
LAKE COUNTY

The California Regional Water Quality Control Board, Central Valley Region, (hereafter Board) finds that:

1. The Board, on 23 October 1998, adopted Waste Discharge Requirements (WDRs) Order No. 98-206 which prescribed requirements for a discharge of treated effluent from the Northwest Regional (NWR) wastewater system for land disposal in Lake County. The NWR Facility is owned and operated by the Lake County Sanitation District (hereafter referred to as "Discharger").
2. Order No. 98-206 prescribed requirements for the operation of a wastewater treatment facility and its discharge of treated effluent to an 865-acre-foot (281 million gallons) effluent storage reservoir. During the dry season, effluent from the reservoir has been sprayed onto 540 acres of pasture land owned by the Discharger.
3. Over the past decade, a lack of effluent storage and disposal capacity resulted in documented overflows from the effluent storage reservoir during periods of above average rainfall. Such overflows have been to Lyons Creek, tributary to Clear Lake, in violation of Order No. 98-206 and Lake County Ordinances. In addition, monthly monitoring reports submitted pursuant to WDRs Order No. 98-206 indicated that the Discharger's NWR treatment facility repeatedly violated effluent limitations for various water quality parameters.
4. Order No. 98-206 is being revised based upon information submitted by the Discharger to facilitate improved wastewater collection, treatment and land disposal practices, and for additional water conveyance and recycling processes through the implementation of the Clear Lake Basin 2000 Project (Project). The Discharger submitted a Report of Waste Discharge (ROWD), dated 6 June 2000, for the proposed changes to its NWR wastewater collection, treatment, recycling and disposal system. Additional documentation, including the July 2000 *Title 22 Engineers Report - Recycled Water Project* (Title 22 Report), the October 1999 *Environmental Assessment/Finding of No Significant Impact - Initial Study/Mitigated Negative Declaration* (Initial Study), letter dated 1 March 2001, facsimile dated 4 April 2001, and numerous design reports were also provided by the Discharger to supplement the ROWD.

5. The Board, on 25 February 1994, adopted Cease and Desist Order (CDO) No. 94-064 containing a time schedule for the Discharger to expand its treatment system and obtain adequate disposal facilities. The Discharger's proposed changes, as reflected in the ROWD and outlined in this Order, are in response to CDO No. 94-064.
6. As indicated in the Title 22 Report, the Project initially consisted of three primary elements:
 - a. Construction of a recycled water pipeline (NWR Pipeline) and two booster pump stations (NWPS Nos. 1 and 2) to deliver recycled water from the existing NWR Reservoir to constructed wetlands sites and to receive the operational discharges from the wetland sites;
 - b. Construction of wetlands to enhance wildlife habitat, recycle treated wastewater, improve the water quality of natural runoff into Clear Lake, and provide public benefits through the creation of passive recreation and wildlife study; and
 - c. Construction of a pipeline connection to the existing Geysers Wastewater Recycling System that uses treated effluent to sustain the geothermal reservoir.

Subsequent to the preparation and circulation of the Title 22 Report, the Project as proposed under this Order has been modified to exclude any additional wetland sites at the NWRWTP, SERWTP and at various locations along the length of the pipeline. As indicated in the Discharger's letter dated 1 March 2001 and contrary to element (a) described above, wetland sites in addition to the demonstration site described in Finding No. 10 of this Order are no longer being considered at this time. The NWR Pipeline and NWPS Nos. 1 and 2 shall provide the means for the deliverance of recycled water through connection to the existing Geysers Wastewater Recycling System.

7. Raw wastewater, primarily of domestic origin, is collected from communities along the northern and western shores of Clear Lake, including Nice, Paradise Valley, Kono Tayee, Lucerne, Upper Lake and North Lakeport. In addition, the treatment facility receives septage from septage haulers, which is dumped in a receiving station at the headworks.
8. The existing wastewater treatment plant, with a design treatment capacity of 2.0 million gallons per day (mgd) and a hydraulic peak capacity of 3.6 mgd, consists of oxidation ditches/secondary clarification with chlorine gas disinfection. Proposed improvements to the treatment system under this Order include conversion of the existing oxidation ditches to fully-mixed lagoon cells. An additional 1.5-million gallon, fully-mixed lagoon cell and approximately 6.8 million gallons of partially mixed lagoons with floating baffles will also be constructed as part of the facility improvements. The Discharger also proposes to switch from chlorine gas to liquid sodium hypochlorite disinfection. Stabilized sludge from the treatment process is currently discharged to a dedicated disposal area on

Discharger-owned land at the treatment plant site. A solids recycling system is proposed for improved biosolids management under this Order.

9. The NWR treatment facility discharges treated effluent to an 865-acre-foot (281 million gallons) effluent storage reservoir (NWR Reservoir). From the NWR Reservoir, effluent may be recycled through spray irrigation to the existing 540-acre recycled water irrigation area, or discharged to a proposed effluent NWR pump station (NWPS No. 1). Upon completion of the Project, the Discharger does not intend to conduct spray irrigation of treated effluent under normal operating conditions. Use of the spray irrigation area will be minimal and typically used on a contingency basis. During periods of recycled water application, all of the excess runoff from the irrigation area is collected in a tailwater control basin and returned to the NWR Reservoir.
10. Treated effluent may also be discharged from the NWR Reservoir via buried pipeline to the upper-most cell in a series of four constructed wetland cells at the NWRWTP (hereafter collectively referred to as "NWR Wetland"). Referred to as NWRWTP/Lyons Creek Wetland Site No. 6 in the ROWD documents, the 5.04-acre NWR Wetland was initially constructed in July 2000 and operated as a pilot study/demonstration project. Intended primarily for wildlife habitat enhancement and restoration, the NWR Wetland is also anticipated to recycle and polish the effluent, manage storm water runoff, reduce peak flood flows, control erosion, and provide public benefits through the creation of settings for passive recreation and wildlife study.
11. As described in a 4 April 2001 facsimile by the Discharger, the operational concept of the NWR Wetland includes gravity flow through the four serially interconnected wetland cells. At the outflow of the final cell, any excess water will be returned to the NWR Reservoir through a recapture pipeline. Storm water management ditches are constructed at the exterior toe of the wetland levees to divert runoff from the surrounding watershed into existing waterways. Any runoff or discharge of recycled water commingled with storm water from the constructed wetlands is to be contained within the Discharger's Designated Recycled Water Areas or returned to the NWR Reservoir.
12. For the purposes of compliance with the California Department of Health Services statewide reclamation criteria established in Title 22, California Code of Regulations, Section 60301, et seq. (hereafter Title 22), the Use Area designation of the NWR Wetland under this Order shall be considered a Landscape Impoundment. Section 60301.550 of Title 22 defines a Landscape Impoundment as "an impoundment in which recycled water is stored or used for aesthetic enjoyment or landscape irrigation, or which otherwise serves a similar function and is not intended to include public contact."

13. The NWR Wetland regulated under this Order is owned by the Discharger and is shown on Attachment A, which is attached hereto and part of the Order by reference.
14. Additional wetland sites at the NWRWTP, SERWTP and at various locations along the length of the pipeline were initially proposed for construction as part of the Project but, as indicated in the Discharger's letter dated 1 March 2001, are no longer being considered at this time. Any future plans for recycled water use in constructed wetlands not located at the NWRWTP are to be covered under separate WDRs or Water Recycling Requirements.
15. The on-site 540-acre recycled water irrigation area and NWR Wetland are referred to as Designated Recycled Water Areas.
16. NWPS No. 1 will also lift recycled water from the NWR Reservoir to an intermediate booster pump station (NWPS No. 2). Equipped with a 150,000-gallon regulating reservoir, NWPS No. 2 is adjacent to the Clearlake Oaks County Water District's (COCWD) recycled water pump station.
17. To facilitate the delivery of recycled water from NWPS No. 1 to NWPS No. 2, a 16-inch pipeline (NWR Pipeline) will extend 21 miles along the west and north shores of Clear Lake with a design flow rate of 4.0 mgd.
18. In October 1999, the COCWD completed construction of a pipeline (COCWD Pipeline) to connect the COCWD's recycled water pump station and NWPS No. 2 to the Discharger's Southeast Regional Wastewater Treatment Plant (SERWTP) storage reservoir. Through this 16-inch diameter, 6,800-foot long COCWD Pipeline, treated and disinfected wastewater from the 2.1 mgd-capacity COCWD pump station and the 4.0 mgd-capacity NWPS No. 2 is delivered to the SERWTP's storage reservoir for ultimate discharge into the Southeast Geysers Pipeline (SEG Pipeline). The COCWD Pipeline is regulated under separate WDRs.
19. The Discharger's SERWTP is regulated under separate WDRs which provides for the discharge of treated effluent from the SERWTP storage reservoir to the SEG Pipeline. The Discharger, in partnership with geothermal steam suppliers and users, constructed the SEG Pipeline to deliver treated and disinfected wastewater combined with supplemental lake water to a geothermal resource area known as KGRA, or the Geysers, for injection. The KGRA is approximately 14 miles southwest of the SERWTP and since the 1970's has been actively tapped for steam production in order to generate electricity. Steam production has decreased over the years and the managed injection of water has proven to prolong the life of the steamfield in specific areas through improved steam reservoir pressures, resulting in increased power production.

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20. The Discharger has the responsibility for the construction, operation, maintenance and monitoring of the two recycled water pump stations (NWPS Nos. 1 and 2), NWR Pipeline, and the NWR Wetland.
21. Wastewater effluent from the NWR treatment system is referred to as effluent. Effluent discharged from the NWR Reservoir to the irrigation area, NWR Wetland, NWR Pipeline and/or COCWD Pipeline is referred to as recycled water.
22. The NWRWTP, NWR Reservoir, 540-acre on-site irrigation disposal area, and NWR Wetland are in Sections 25, 26, 35 and 36, T15N, R10W, MDB&M (Assessor's Parcel No. 003-046-30).
23. The California Department of Water Resources has established standards for the construction and destruction of groundwater wells. These standards are described in two DWR publications: *California Well Standards Bulletin 74-90* (June 1991) and *Water Well Standards: State of California Bulletin 94-81* (December 1981).
24. The Basin Plan encourages reclamation. Injection of blended effluent and lake water for steam/electric production is an encouraged reclamation project.
25. The California Department of Health Services has established statewide reclamation criteria in Title 22, California Code of Regulations, Section 60301, et seq. (hereafter Title 22) for the use of reclaimed water and has developed guidelines for specific uses.
26. The Discharger submitted an engineering report, dated July 2000, in accordance with the California Code of Regulations (CCR) Title 22, Division 4, Chapter 3, Article 7, Section 60323, Water Reclamation Criteria.
27. The Board consulted with the Department of Health Services, Lake County Health Department, and Mosquito Abatement District and considered their recommendations regarding public health aspects for use of recycled water.
28. The *Water Quality Control Plan for the Sacramento River and San Joaquin River Basins, Fourth Edition*, (hereafter Basin Plan), designates beneficial uses, establishes water quality objectives, contains implementation plans and policies for protecting waters of the basin, and incorporates by references plans and policies adopted by the State Water Resources Control Board. These requirements implement the Basin Plan.
29. Pursuant to CWC section 13263(g), discharge is a privilege, not a right and issuance of this Order does not create a vested right to continue the discharge. Failure to provide the level of management required to assure best practicable treatment and control, preclude

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conditions that threaten pollution or nuisance, and maintain the highest quality groundwater will be sufficient reason to modify this Order, revoke the Order, or enforce the Order and prohibit further discharge.

30. Section 13267(b) of the California Water Code provides that: "In conducting an investigation specified in subdivision (a), the regional board may require that any person who has discharged, discharges, or is suspected of discharging, or who proposes to discharge waste within its region, or any citizen or domiciliary, or political agency or entity of this state who has discharged, discharges, or is suspected of discharging, or who proposes to discharge waste outside of its region that could affect the quality of waters of the state within its region shall furnish, under penalty of perjury, technical or monitoring program reports which the regional board requires. The burden, including costs, of these reports shall bear a reasonable relationship to the need for the report and the benefits to be obtained from the reports."
31. The Board has considered antidegradation pursuant to State Board Resolution No. 68-16 and finds that the conditional discharge as permitted herein is consistent with those provisions. If the discharge threatens to cause or causes an impact to groundwater quality, the Discharger will be required to cease the discharge, implement source control, alter the method of disposal, or take other actions as necessary to prevent degradation.
32. Surface water drainage is to Lyons Creek, which is tributary to Clear Lake, thence to Cache Creek.
33. The beneficial uses of Lyons Creek, Clear Lake and Cache Creek are municipal, industrial and agricultural supply; recreation; aesthetic enjoyment; navigation; ground water recharge; fresh water replenishment; hydroelectric power generation; and preservation and enhancement of fish, wildlife, and other aquatic resources.
34. The beneficial uses of underlying groundwater are domestic, municipal, industrial, and agricultural supply.
35. In August 1997, the Discharger's Board of Directors certified the *Northwest Regional Wastewater Treatment and Disposal Facility Improvements EIR* (EIR), which examined proposed improvements for collection system, treatment systems, recycled effluent storage, and reuse. Subsequent to approval of the abovementioned EIR, the Discharger proposed additional wetland sites for implementation along the State Route 20 corridor that were not previously identified in the analysis. Therefore, on 2 October 1999 the U.S. Environmental Protection Agency adopted an October 1999 *Clear Lake Basin 2000 - Phase II Environmental Assessment/Finding of No Significant Impact and Initial Study/Mitigated Negative Declaration* in accordance with the National Environmental

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Policy Act (NEPA). An Initial Study/Mitigated Negative Declaration was approved by the Discharger's Board of Directors on 7 February 2000 in accordance with the California Environmental Quality Act (CEQA) (Public Resources Code Section 21000, et. seq.) and the State CEQA Guidelines. As stated in the Mitigated Negative Declaration, the project would not have a significant effect on the environment, because mitigation measures have been incorporated into the proposed project.

36. This discharge is exempt from the requirements of *Consolidated Regulations for Treatment, Storage, Processing, or Disposal of Solid Waste*, as set forth in Title 27, CCR, Division 2, Subdivision 1, Section 20005, et seq., (hereafter Title 27). The exemption, pursuant to Section 20090(b), is based on the following:
 - a. The Board is issuing Waste Discharge Requirements,
 - b. The discharge complies with the Basin Plan, and
 - c. The wastewater does not need to be managed according to 22 CCR, Division 4.5, Chapter 11, as a hazardous waste.
37. The Board has notified the Discharger and interested agencies and persons of its intent to prescribe Waste Discharge Requirements for this discharge and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.
38. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that Order No. 98-206 is rescinded and Lake County Sanitation District, Northwest Regional Wastewater System, its agents, successors, and assigns, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following:

A. Discharge Prohibitions:

1. Discharge of wastes, effluent, recycled water or injection water to surface waters or surface water drainage courses is prohibited.
2. There shall be no bypassing of untreated or partially treated wastewater from the wastewater treatment plant or any intermediate unit processes to the point of use.
3. Discharges or releases of recycled water from any segment of the NWR Pipeline, other than into the COCWD Pipeline, is prohibited.

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4. Application or impoundment of wastewater or recycled water within 100 feet of any well used for domestic water supply is prohibited.
5. The discharge of waste classified as 'hazardous', as defined in Sections 2521(a) of CCR, Title 23, Section 2510, et seq. (hereafter Chapter 15), or 'designated', as defined in Section 13173 of the California Water Code, is prohibited.
6. The application of effluent or recycled water on areas which do not have vegetative cover or other erosion control measures in effect is prohibited.

B. Discharge Specifications:

1. The discharge or use of treated effluent shall not cause the degradation of surface or groundwater.
2. The 30-day average daily dry weather effluent discharge flow from the NWRWTP to the NWR Reservoir shall not exceed 2.0 mgd.
3. Neither the collection, treatment, storage, recycling or discharge shall cause a nuisance or condition of pollution as defined by California Water Code, Section 13050.
4. Objectionable odors originating at the Discharger's facility shall not be perceivable beyond the limits of the wastewater treatment, recycling and disposal areas (property owned by the Discharger).
5. The Discharger shall initiate and document an odor investigation within 24 hours upon receiving complaints of odor production.
6. As a means of discerning compliance with Discharge Specification No. 4, the dissolved oxygen content in the upper zone (1 foot) of wastewater in the NWR effluent storage Reservoir and NWR Wetland shall not be less than 1.0 mg/L.
7. Process or equipment failures triggering an alarm shall be recorded and maintained as a record file. The recorded information shall include the time and cause of failure and corrective action taken.
8. The NWR Reservoir and constructed wetlands shall not have a pH less than 6.5 or greater than 9.0 in the upper zone (1 foot) of the reservoir or wetland.
9. A minimum 2.0 feet of freeboard, as measured from the lowest portion of the levee or spillway at which overflow is likely to occur, shall be maintained in the NWR Reservoir at all times.

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10. A minimum 1.0 foot of freeboard, as measured from the lowest portion of the levee at which overflow is likely to occur, shall be maintained in the final cell (cell No. 4) of the NWR Wetland at all times.
11. The NWR Reservoir, NWR Wetland and NWPS No. 2 regulating reservoir shall have sufficient capacity, in conjunction with disposal capacity, to accommodate allowable wastewater flow, design seasonal precipitation and ancillary inflow and infiltration. Design seasonal precipitation shall be based on total annual precipitation using a return period of 100 years, distributed monthly in accordance with historical rainfall patterns.
12. On or about 1 **October** of each year, available reservoir storage capacity shall at least equal the volume necessary to comply with Discharge Specification Nos. 9, 10 and 11.
13. Microbiological (Total and Fecal Coliform) sampling and analysis shall be performed in such a way as to yield sufficient information regarding actual coliform concentrations. A test result of >1600 MPN/100 mL does not provide sufficient information to characterize a wastewater discharge or groundwater infiltration. If results are anticipated to be >1600 MPN/100 mL, or analyzed as such, the Discharger shall immediately re-sample and specify that the analysis be carried out until termination.
14. Effluent from the NWRWTP shall not exceed the following limits:

<u>Constituent</u>	<u>Units</u>	<u>30-day Average</u>	<u>7-Day Median</u>	<u>Daily Maximum</u>
BOD ₅ ¹	mg/l	40		80
Settleable Solids	ml/l	0.2		1.0
Total Coliform ²	MPN/100ml		23	240

¹ Five-day, 20° Celsius biochemical oxygen demand (BOD).

² Total Coliform test(s) must be of sufficient dilution to provide actual coliform concentrations.

15. Discharge of recycled water shall only occur to constructed wetland sites owned by the Discharger and regulated as Designated Water Recycling Areas under this Order.
16. Recycled water discharge shall remain within the Designated Water Recycling Areas at all times.

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17. Recycled water used for irrigation or discharge to constructed wetlands shall meet the criteria contained in Title 22, CCR, Division 4, Section 60301, et seq.
18. All treated effluent, storage, recycling and disposal equipment, pumps, piping, valves, and outlets shall be appropriately marked and tagged to differentiate them from potable facilities.
19. No physical connection shall be made or allowed to exist between the recycled water system and any separate system conveying potable water.
20. The portion of the recycled water piping system in areas subject to access by the general public shall not include any hose bibbs.
21. Public contact with effluent or recycled water shall be precluded through such means as signs, fences, or other acceptable alternatives approved by the Executive Officer.
22. Recycled water irrigation shall not occur within 100 feet of any area accessible to the public.
23. Irrigation or impoundment of effluent or recycled water shall not occur within 100 feet of any domestic water supply well.
24. The application of recycled water to irrigation areas is prohibited during periods of precipitation and for 24 hours after cessation of precipitation.
25. Application of recycled water to irrigation areas shall not result in runoff or discharge to surface water drainage courses, including Lyon's Creek.
26. All runoff (including storm water) from recycled water irrigation areas shall be returned to the NWR Reservoir or appropriate irrigation block for a minimum of 48 hours following any recycled water irrigation.
27. Any discharge of recycled water or storm water from within the constructed wetland cells shall be contained within the Designated Recycled Water Areas or returned to the NWR Reservoir.
28. There shall be no standing recycled water in the recycled water irrigation area 24 hours after recycled water is applied.
29. Recycled water used for irrigation shall be managed to minimize erosion, runoff, and movement of aerosols from the disposal area.

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30. The NWR Pipeline shall be constructed such that the alignment and locations satisfy the minimum horizontal and vertical separation distances prescribed by Title 22, CCR, Division 4, Chapter 16, Article 5, California Water Works Standards, Section 64630(c), (d) and (e).
31. The wastewater treatment, storage, and disposal facilities shall be managed to prevent breeding of mosquitos. In particular,
 - a. An erosion control program should assure that small coves and irregularities are not created around the perimeter of the water surface.
 - b. Weeds shall be minimized through control of water depth, harvesting or herbicides.
 - c. Dead algae, vegetation and debris shall not accumulate on the water surface.
32. Constructed wetland areas shall be managed to prevent breeding of mosquitos as described in the plan required in Provision E.13 and approved by the Executive Officer and local Mosquito Abatement district.
33. The injection of chemicals or additives to recycled water within the NWR Wetland (i.e.: for odor control, etc.) shall not affect the operation of the constructed wetlands and shall be non-toxic and biodegradable in nature.

C. Sludge Disposal:

1. Collected screenings, sludges, and other solids removed from liquid wastes shall be disposed of in a manner that is consistent with Title 27 and approved by the Executive Officer.
2. Any proposed change in sludge use or disposal practice from a previously approved practice shall be reported to the Executive Officer and U.S. Environmental Protection Agency (EPA) Regional Administrator at least 90 days in advance of the change.
3. Use and disposal of sludge shall comply with existing Federal and State laws and regulations, including permitting requirements and technical standards included in 40 CFR 503.

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If the State Water Resources Control Board and the Regional Water Quality Control Boards are given the authority to implement regulations contained in 40 CFR 503, this Order may be reopened to incorporate appropriate time schedules and technical standards. The Discharger must comply with the standards and time schedules contained in 40 CFR 503 whether or not they have been incorporated into this Order.

4. The Discharger is encouraged to comply with the State Guidance Manual issued by the Department of Health Services titled *Manual of Good Practice for Landspreading of Sewage Sludge*.
5. By **15 August 2002**, the Discharger shall submit an updated sludge disposal plan describing the annual volume of sludge generated by the plant and specifying disposal practices consistent with the California Code of Regulations and the Code of Federal Regulations.

D. Groundwater Limitations:

The discharge, in combination with other sources, shall not cause underlying groundwater to contain waste constituents in concentrations statistically greater than background water quality, except for coliform. For coliform, increases shall not cause the most probable number of total coliform organisms to exceed 2.2/100 ml over any 7-day period.

E. Provisions:

1. The Discharger shall comply with the Monitoring and Reporting Program No. 5-01-111, which is part of this Order, and any revisions thereto as ordered by the Executive Officer.
2. The Discharger shall comply with the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements", dated 1 March 1991, which are attached hereto and by reference a part of this Order. This attachment and its individual paragraphs are commonly referenced as "Standard Provision(s)."
3. The Discharger shall not allow pollutant-free wastewater to be discharged into the collection, treatment, and disposal system in amounts that significantly diminish the system's capability to comply with this Order. Pollutant-free wastewater means rainfall, groundwater, surface water, cooling waters, and condensates that are essentially free of pollutants.

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4. By the effective date of this Order, the Discharger shall provide certified wastewater treatment plant operators in accordance with regulations adopted by the State Water Resources Control Board.
5. In the event of any change in control or ownership of land or waste discharge facilities described herein (pipeline and wetland facilities), the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be immediately forwarded to this office.
6. At least **90 days** prior to termination or expiration of any lease, contract, or agreement involving disposal or water recycling areas or off-site reuse of effluent, used to justify the capacity authorized herein and assure compliance with this Order, the Discharger shall notify the Board in writing of the situation and of what measures have been taken or are being taken to assure full compliance with this Order.
7. The Discharger shall submit a Report of Waste Discharge and request for approval from the Executive Officer at least **180 days prior** to any intended discharge of recycled water to a third party user.
8. The Discharger must comply with all conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Violations may result in enforcement action, including Regional Board or court orders requiring corrective action or imposing civil monetary liability, or in revision or rescission of this Order.
9. The Discharger shall submit a spill response and emergency/contingency plan at least **90 days** prior to the operation of the NWR Pipeline to assure compliance with this Order.
10. By **15 November 2001**, the Discharger shall submit a revised hydraulic water balance for the Project incorporating anticipated reductions in available recycled water irrigation area and only considering the use of wetland sites regulated under this Order. The revised water balance shall be in accordance with the minimum requirements identified in Discharge Specification No. B.11.
11. By **1 December 2001**, the Discharger shall submit for Executive Officer approval, a technical report proposing a groundwater monitoring network. The technical report shall also contain a Monitoring Well Installation Workplan and implementation schedule. The network shall consist of one or more background monitoring wells

and of compliance monitoring wells immediately downgradient of the NWRWTP, NWR Reservoir and the NWR Wetland to define ground water impacts from each discharge area. The number and location of wells downgradient of the facilities shall be sufficient to determine ground water impacts. Every monitoring well shall be constructed to yield representative samples from the uppermost layer of the uppermost aquifer and comply with applicable Well Standards. Implementation of the Monitoring Well Installation Workplan shall be subject to the prior approval of the Executive Officer. Groundwater monitoring shall commence **within 90 days of Executive Officer approval of the Installation Workplan.**

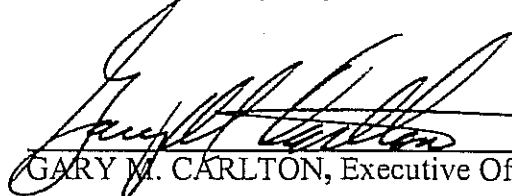
Technical reports submitted pursuant to this Provision shall be prepared and certified by a California registered civil engineer or registered geologist.

12. By **1 January 2002**, the Discharger shall complete necessary facility modifications so that recycled water discharge (including storm water) from the NWR Wetland does not enter into any surface water course, including the recapture pond located in the Lyon's Creek drainage course, prior to returning to the NWR Reservoir.
13. By **1 January 2003**, the Discharger shall submit an Operations and Maintenance (O&M) Plan for management of the NWRWTP, Designated Water Recycling Areas and NWR Pipeline. At a minimum, the O&M Plan shall identify varying levels of staff responsibility, methods for conducting routine monitoring, description of process control features and alarms, irrigation area and wetland management, hydraulic flow regimes, nuisance prevention practices (odors, algal blooms, etc.), wetland vegetation maintenance and control, security and public access issues, and responsibilities and coordination with COCWD for operation and maintenance of the COCWD Pipeline.
14. The Discharger shall submit a plan and schedule for the management and control of mosquitos and vectors at the constructed wetland sites regulated under this Order. The plan shall be submitted for approval to the local Mosquito Abatement district and the Executive Officer of the Central Valley Regional Water Quality Control Board by **1 September 2001**.
15. A copy of this Order shall be kept at the discharge facility for reference by operating personnel. Key operating personnel shall be familiar with its contents.
16. The Board will review this Order periodically and will revise requirements when necessary.

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I, GARY M. CARLTON, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 11 May 2001.



GARY M. CARLTON, Executive Officer

SKC: 5/11/01

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

MONITORING AND REPORTING PROGRAM NO. 5-01-111

FOR
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The Discharger shall comply with the following Monitoring and Reporting Program (MRP), issued pursuant to Water Code Section 13267, which describes requirements for monitoring wastewater, recycled water, soil, surface water and groundwater. The Discharger shall not implement any changes to this MRP unless and until a revised MRP is issued by the Executive Officer.

All samples should be representative of the volume and nature of the discharge or matrix of material sampled. The time, date, and location of each grab sample shall be recorded on sample chain of custody forms.

Copies of laboratory analytical reports shall be maintained onsite for a minimum of five years and be available for review by Regional Board staff.

INFLUENT MONITORING

Samples shall be collected at approximately the same time as effluent samples and should be representative of the influent for the period sampled. The following shall constitute the influent monitoring program:

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>
Flow	gpd	Continuous	Daily
pH ¹	pH Units	Grab	Weekly
Specific Conductance ¹	µmhos/cm	Grab	Weekly
BOD ₅ ²	mg/l	24 hr. composite	Weekly

¹ A hand held field unit may be used to measure specific conductance and pH.

² 5-day, 20°

C Biological Oxygen Demand.

TREATMENT PLANT EFFLUENT MONITORING

Treatment plant effluent samples shall be collected from the wastewater treatment facility just prior to discharge to the NWR Reservoir. Effluent samples should be representative of the volume and nature of the discharge. Samples collected from the overflow weir from the treatment plant to the NWR effluent storage reservoir will be considered adequately composited. Time of collection of a grab sample shall be recorded. Treatment plant effluent monitoring shall include at least the following:

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<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>
Total Coliform Organisms	MPN/100 ml	Grab	Daily
pH ¹	pH Units	Grab	Weekly
Specific Conductance ¹	µmhos/cm	Grab	Weekly
BOD ₅ ²	mg/l	24 hr. composite	Weekly
Settleable Solids	ml/l	Grab	Weekly
Freeboard ³	feet (±0.5)	Measured	Weekly
Chlorine Residual	mg/l	Grab	Weekly
Total Dissolved Solids	mg/l	Grab	Monthly
Nitrates as Nitrogen	mg/l	Grab	Monthly

¹ A hand held field unit may be used to measure specific conductance and pH.

² 5-day, 20°C Biological Oxygen Demand.

³ The vertical distance between the surface of the NWR Reservoir to the base of the reservoir spillway. Permanent markers shall be placed in the NWR Reservoir with calibration indicating the water level at design capacity and available operational freeboard.

SLUDGE MONITORING

The Discharger shall monitor and report the sludge production and quality in accordance with the sludge production and disposal plan approved by the Executive Officer.

RECYCLED WATER IRRIGATION AREA MONITORING

Inspections of the recycled water irrigation areas shall be conducted daily during periods of irrigation; the observations shall be included in the monitoring reports. Evidence of erosion, field saturation, ponding, runoff, accumulation of solids, soil clogging, presence of nuisance such as odors, vectors or insects, and other relevant field conditions shall be observed, recorded, and reported in the regular monitoring reports. The notations shall also document any corrective actions taken with respect to observations made. A copy of the entries made in the log during each month shall be submitted along with monthly monitoring reports.

Daily discharge volumes and total acres irrigated shall be noted in the monitoring report. Specific dates of irrigation and the cumulative number of days irrigated in the previous 12 months shall also be recorded and submitted with each monthly monitoring report. The report shall also include a discussion of the discharge volumes and irrigation practices used (method of application, application period/duration, drying times, etc.) for the various portions of the recycled water irrigation areas throughout the month. Hydraulic loading rates (in/acre/month) shall be calculated on a monthly basis during periods of application.

SOIL MONITORING

Soil monitoring is needed to ascertain whether the nitrate-nitrogen loading is in excess of that required by the vegetation. A minimum of two composite soil samples shall be collected from the recycled water irrigation area. Each composite sample shall be comprised of four to ten discrete soil samples collected throughout the recycled water irrigation area. If the recycled water irrigation area has been used for effluent disposal on greater than 20 days during any consecutive 12 month period, soil samples shall be collected from the spray irrigation area at a depth of one foot and analyzed for the following compounds:

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>
Salinity	mg/kg	Composite	Annual ¹
Nitrogen	mg/kg	Composite	Annual ¹
Phosphorus	mg/kg	Composite	Annual ¹
Potassium	mg/kg	Composite	Annual ¹

¹ Samples shall be collected prior to the onset of the rainy season for only those periods in which discharge occurs.

RECYCLED WATER PIPELINE MONITORING

Samples shall be collected at the point discharge of effluent and/or recycled water from the NWR Reservoir to the NWR Pipeline (samples collected from the NWPS No. 1 will be considered adequately composited) as follows:

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>
Flow ¹	gpd	Continuous	Daily
pH ^{2,3}	pH Units	Grab	Weekly
Specific Conductance ²	µmhos/cm	Grab	Weekly
Settleable Solids	ml/l	Grab	Weekly
Total Coliform Organisms	MPN/100 ml	Grab	Monthly
BOD ₅ ⁴	mg/l	Grab	Monthly
Total Dissolved Solids	mg/L	Grab	Monthly
Nitrates as Nitrogen	mg/L	Grab	Monthly

¹ Flow shall also be measured from the NWPS No. 1 into the NWR Pipeline.

² A hand held field unit may be used to measure specific conductance and pH.

³ If the compound is outside the permissible limits established in Discharge Specification Nos. B.6 and B.8 of WDRs Order No. 5-01-111, the Discharger shall resample within the designated sampling frequency period to determine if compliance was achieved for the monitoring period.

⁴ 5-day, 20°C Biological Oxygen Demand.

CONSTRUCTED WETLAND MONITORING

Daily inspections of each cell in the NWR Wetland shall be conducted and recorded by the Discharger to confirm that the inlet and outlet structures are operating as intended. Observations of wetland conditions such as color, odor, vegetative growth and public safety/security provisions shall also be recorded on a weekly basis.

Wetland recycled water monitoring shall also consist of the following of both the influent and effluent from the NWR Wetland:-

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency¹</u>
Flow	gpd	Continuous	Daily
Freeboard in final cell	feet (+0.1 ft)	Measurement	Weekly
pH ^{2,3}	pH Units	Grab	Weekly
Specific Conductance ²	µmhos/cm	Grab	Weekly
Dissolved Oxygen ^{2,3}	mg/l	Grab	Monthly
BOD ⁴	mg/l	Grab	Quarterly
Settleable Solids	ml/l	Grab	Quarterly
Total Dissolved Solids	mg/l	Grab	Quarterly
Nitrates as Nitrogen	mg/l	Grab	Quarterly
Total Coliform Organisms	MPN/100 ml	Grab	Quarterly

¹ Inlet and outlet samples shall be collected on the same day within the specified monitoring period. The monitoring frequency may be reduced upon written request from the Discharger and supported by proven performance data. Requests for reduction in monitoring must address the requirements of the Monitoring Reduction Request Report, as defined under the Reporting section below. Reductions in monitoring requirements shall not occur until approved by the Executive Officer.

² A hand held field unit may be used to measure dissolved oxygen, specific conductance and pH.

³ If the compound is outside the permissible limits established in Discharge Specification Nos. B.6 and B.8 of WDRs Order No. 5-01-111, the Discharger shall resample within the designated sampling frequency period to determine if compliance was achieved for the monitoring period.

⁴ 5-day, 20°C Biological Oxygen Demand.

The Discharger shall summarize and report any additional water quality monitoring or wetland assessments conducted at the NWR Wetlands in each monthly monitoring report submittal.

GROUNDWATER MONITORING

The Discharger shall develop a network of monitoring wells at the NWRWTP and Designated Water Recycling Areas to define ground water impacts from the NWRWTP, NWR Reservoir, and Designated Recycled Water Areas. At least one well shall be established upgradient of the facilities to monitor

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background levels. The number and location of wells downgradient of the facilities shall be sufficient to determine ground water impacts. As required by Provision No. 11, the Discharger shall submit a technical report with a plan and schedule for installation of the monitoring well systems for approval by the Executive Officer by **1 December 2001**. At a minimum, the approved monitoring wells shall be sampled as follows:

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency^{1,2}</u>
Groundwater Elevation ³	Feet (± 0.1)	Measurement	Quarterly
pH	pH Units	Grab	Quarterly
Specific Conductance	$\mu\text{mhos/cm}$	Grab	Quarterly
Total Dissolved Solids	mg/L	Grab	Quarterly
Nitrates as Nitrogen	mg/l	Grab	Quarterly
Total Coliform Organisms	MPN/100 ml	Grab	Quarterly
General Minerals	mg/l	Grab	Semi-Annually ^{2,4,5}

¹ The Discharger may propose a reduction in frequency to semi-annual provided the data demonstrate no significant changes in water quality. Requests for reduction in monitoring must address the requirements of the Monitoring Reduction Request Report, as defined under the Reporting section below. Reductions in monitoring requirements shall not occur until approved by the Executive Officer.

² Semi-annual monitoring shall be performed at the height of the wet (typically January/February) and dry (typically August/September) seasons.

³ Groundwater elevation shall be determined based on depth-to-water measurements from a surveyed measuring point elevation on the well.

⁴ A "fingerprint" of the groundwater, for characterization purposes, shall be conducted for two years.

⁵ General Minerals shall include at least the following compounds: calcium, potassium, sodium, chloride, nitrate, sulfate, total dissolved solids, electrical conductivity, pH, total alkalinity, and total hardness.

Prior to collecting samples, monitoring wells shall be adequately purged to remove water that has been standing within the well screen and casing that may not be chemically representative of formation water. Depending on the hydraulic conductivity of the geologic setting, the volume removed during purging is typically from 3 to 5 volumes of the standing water within the well casing and screen (or additionally the filter pack pore volume), or until stabilization of temperature, pH and EC, or refilling of the well after dewatering is achieved.

Prior to monitoring well purging and sampling, the Discharger shall measure the water level in each well. The Discharger shall report water level data as groundwater depth (in feet and hundredths) and as groundwater surface elevation (in feet and hundredths above mean sea level). The Discharger shall include in its quarterly submittal of groundwater elevation data, a contour map based on said data showing the gradient and direction of groundwater flow under/around the NWRWTP, NWR Reservoir and Designated Recycled Water Areas.

After one full year of groundwater monitoring, the Discharger shall analyze monitoring data from background well(s) to compute background water quality values for each waste constituent and to perform an initial assessment of whether there is evidence of an impact from the discharge. To complete this task, the Discharger shall use monitoring data from background and compliance monitoring wells in an appropriate data analysis method as described in Title 27, Section 20415(e)(7-9). Reports thereafter shall be submitted quarterly by the 1st day of the second month after the prescribed sample collection and shall include the same analysis.

If the Discharger during any quarterly data evaluation confirms statistically significant evidence of an increase in a waste constituent in groundwater at monitoring wells compared to background levels (or other groundwater limitations), the Discharger shall conclude that it is in violation of waste discharge requirements unless it can demonstrate an offsite source. The Discharger shall submit confirmation of violation in a written report pursuant to Standard Provision B.1.

SURFACE WATER MONITORING

A visual observation of the Discharger's NWRWTP, NWR Reservoir and Designated Recycled Water Areas shall be made each day during normal working hours to determine if there is any runoff from the confines of the Discharger's property to surface drainage courses. A log shall be maintained as follows - discharge to surface drainage courses: Yes or No. If a discharge occurs, immediate action consistent with the Discharger's Contingency/Emergency Response Plan, approved by the Executive Officer, shall take place.

Surface water samples shall be routinely collected from Lyons Creek. These samples are needed to assess whether groundwater quality data is influenced by the presence of flow in the creek and/or impacts from the NWR Reservoir and/or Designated Recycled Water Areas. Surface water samples shall be collected at a minimum of two points, (at least one upstream and one downstream of the NWR Reservoir and/or Designated Recycled Water Areas), when flowing, for the following:

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u> ¹
Flow	Yes/No	Observation	Monthly
pH ²	pH Units	Grab ³	Quarterly
Specific Conductance ²	µmhos/cm	Grab ³	Quarterly

¹ The Discharger may propose a reduction in frequency to semi-annual provided the data demonstrate no significant changes in water quality. Requests for reduction in monitoring must address the requirements of the Monitoring Reduction Request Report, as defined under the **Reporting** section below. Reductions in monitoring requirements shall not occur until approved by the Executive Officer.

² A hand held field unit may be used to measure specific conductance and pH.

³ Samples shall be collected when there is flow in the creek and shall be collected contemporaneously with the groundwater monitoring event.

REPORTING

In reporting monitoring data, the Discharger shall arrange the data in tabular form so that the date, sample type (e.g. constructed wetland, groundwater well, etc.) and the reported analytical result for each sample are readily discernible. The data shall be summarized in such a manner to clearly illustrate compliance with waste discharge requirements and spatial or temporal trends, as applicable. The results of any monitoring done more frequently than required at the locations specified in the Monitoring and Reporting Program shall be reported in the next scheduled monitoring report.

As required by the California Business and Professions Code Sections 6735, 7835, and 7835.1, all groundwater monitoring reports shall be prepared under the direct supervision of a registered professional engineer or geologist and signed by the registered professional.

A. Monthly Monitoring Reports

Monthly reports shall be submitted to the Regional Board by the 20th day of the month following sampling (i.e. the January Report is due by 20 February). Monthly reports for the months of March, June, September, and December may be submitted as part of the Quarterly Monitoring Report. Monthly monitoring reports shall contain all monitoring data required as described above.

B. Quarterly Monitoring Reports

Quarterly Monitoring Reports shall be submitted to the Regional Board by the 1st day of February (4th Quarter Report), May (1st Quarter Report), August (2nd Quarter Report), and November (3rd Quarter Report). The Quarterly Reports shall include the results of all regular monthly monitoring data generated and the results of groundwater and surface water monitoring performed. At a minimum, the report shall contain:

1. A narrative description of all preparatory, monitoring, sampling, and analytical testing activities. The narrative shall be sufficiently detailed to verify compliance with the WDRs, this MRP, and the Standard Provisions and Reporting Requirements. Field logs shall support the narrative for each well documenting depth to groundwater; parameters measured before, during, and after purging; method of purging; calculation of the casing volume; and total volume of water purged.
2. Calculation of groundwater elevations, an assessment of the groundwater flow direction and gradient on the date of measurement, comparison to previous flow direction and gradient data, and discussion of seasonal trends, if any.

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3. A narrative discussion of the analytical results for all media and locations monitored, including spatial and temporal trends, with reference to summary data tables, graphs, and appended analytical reports (as applicable).
4. A comparison of monitoring data to the discharge specifications, groundwater limitations, and surface water limitations, and explanation of any violation of those requirements.
5. Summary data tables of historical and current water table elevations and analytical results.
6. A scaled map showing relevant structures and features of the facility, the locations of monitoring wells and other sampling stations, and groundwater elevation contours referenced to mean sea level datum.
7. If requested by staff, copies of laboratory analytical report(s).

C. Annual Report

An Annual Report shall be prepared as the fourth quarter monitoring report. The Annual Report will include all monitoring data required in the monthly/quarterly schedule. The Annual Report shall be submitted to the Regional Board by **15 March** of each year. In addition to the data normally presented, the Annual Report shall include the following:

1. If requested by Staff, tabular and graphical summaries of all data collected during the year.
2. An evaluation of the performance of the constructed wetlands, as well as a forecast of the flows anticipated to be received by the NWR wastewater system in the next year.
3. Summary of total monthly recycled water application, total recycled water applied annually, and total annual nutrient loading from the applied wastewater. Discharge volumes shall be presented in terms of gallons and as inches. The report shall present a mass balance relative to constituents of concern and hydraulic loading along with supporting data and calculations. The report shall describe the types of vegetation/crops at the Recycled Water Irrigation Area, and the vegetation/crop nitrogen content (if determined). The report shall also include a comprehensive evaluation of the effectiveness of the past year's wastewater application operation in protecting groundwater quality. This shall include an overall summary of application management practices (i.e.: waste constituent and hydraulic loadings, drying times, and cropping practices), soil profile monitoring data and groundwater monitoring data.
4. A summary data table of water levels and freeboard in the NWR effluent storage reservoir and constructed wetlands.

5. A discussion of compliance and the corrective action taken, as well as any planned or proposed actions needed to bring the discharge into full compliance with the waste discharge requirements.
6. A discussion of any data gaps and potential deficiencies/redundancies in the monitoring system or reporting program.

D. Monitoring Reduction Request Report

In the event the Discharger believes that a reduction in the monitoring requirements of this Order are appropriate, a summary of the analytical results and any other required measurements must be submitted to justify the proposed changes. The purpose of this report is to identify analytical results or other measurements that are redundant, excessive, or represent steady-state conditions, thereby streamlining the monitoring, reporting and review process, and releasing resources for any needed waste management improvements. Board staff will review the report to concur or refute the proposed recommendations for changes in the monitoring program. Reductions in monitoring requirements shall not occur until approved by the Executive Officer. At a minimum a monitoring reduction report shall contain the following:

1. A scaled map showing relevant structures and features of the facility and the locations and any other sampling stations;
2. Tabular summaries for the historical period or for the segment of the relevant data record;
3. Trend graphs for the historical period or for the segment of the relevant data record; and
4. Any maps, particularly for groundwater (i.e. groundwater elevation flow maps), which will assist in the demonstration of the consistency of the data record.

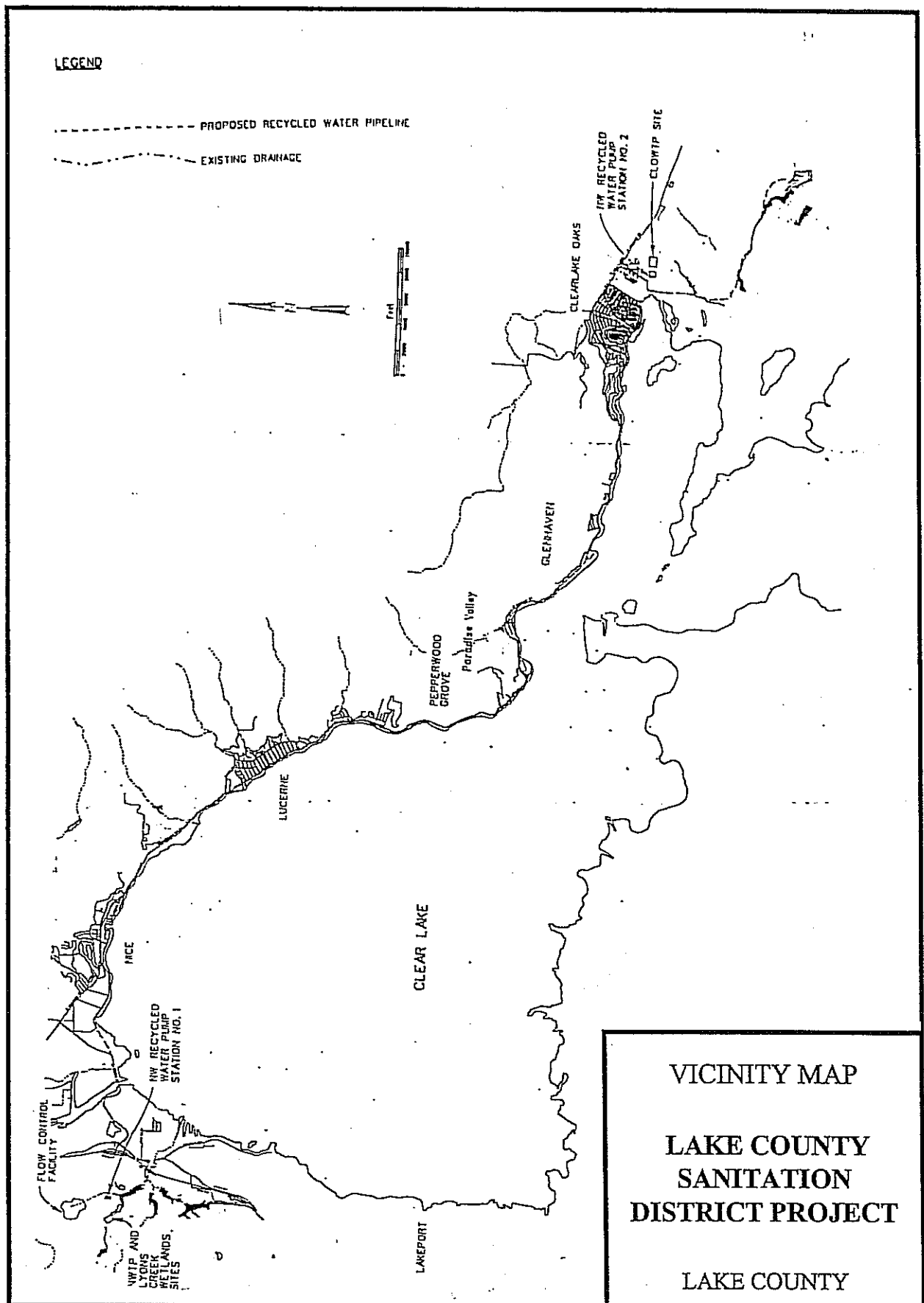
The Discharger shall implement the above monitoring program as of the date of this Order.

Ordered by: _____

GARY M. CARLTON, Executive Officer

11 May 2001

(Date)



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The Northwest Regional (NWR) Wastewater System is owned and operated by the Lake County Sanitation District (hereafter referred to as "Discharger"). The system serves communities along the northern and western shores of Clear Lake, including Nice, Paradise Valley, Kono Tayee, Lucerne, Upper Lake and North Lakeport. Source control programs are not in place as raw wastewater and septage collected from these communities are primarily of domestic or resort-related commercial origin.

Waste Discharge Requirements (WDRs) were first set forth in Order No. 74-050 and subsequently revised in 1985, 1988 and 1998. WDRs Order No. 98-206, adopted by the Board on 23 October 1998, is being revised to implement the Discharger's proposed treatment plant upgrades and the Clear Lake Basin 2000 Project (Project). The Project was proposed by the Discharger for improved wastewater collection, treatment and land disposal practices, and for additional water conveyance and recycling processes. As indicated in the Discharger's July 2000 Title 22 Engineers Report, Recycled Water Project, the Project initially consisted of three primary elements:

- a. Construction of a recycled water pipeline (NWR Pipeline) and two booster pump stations (NWPS Nos. 1 and 2) to deliver recycled water from the existing NWR Reservoir to constructed wetlands sites and to receive the operational discharges from the wetland sites;
- b. Construction of wetlands to enhance wildlife habitat, recycle treated wastewater, improve the water quality of natural runoff into Clear Lake, and provide public benefits through the creation of passive recreation and wildlife study; and
- c. Construction of a pipeline connection to the existing Geysers Wastewater Recycling System that uses treated effluent to sustain the geothermal reservoir.

Contrary to element (a) described above, revisions to the Project have been proposed by the Discharger in a letter dated 1 March 2001, resulting in a reduction in the number of proposed constructed wetlands to be considered by this Order.

WDRs Order No. 98-206 prescribed requirements for the operation of a wastewater treatment facility that discharged its treated effluent to an 865 acre-foot (281 million gallons) storage reservoir. Under WDRs Order No. 98-206, the Northwest Regional Wastewater Treatment Plant (NWRWTP) provided secondary treatment using oxidation ditches/secondary clarifiers, chlorine disinfection and sludge drying beds. Stabilized sludge from the treatment process was discharged to a dedicated storage/disposal area at the treatment plant site. During the dry season, effluent from the reservoir was sprayed and recycled onto 540 acres of pastureland owned by the Discharger.

Prior to the completion of the Project improvements, the NWRWTP had a design treatment capacity of 2.0 million gallons per day (mgd) and a hydraulic peak capacity of 3.6 mgd. Over the last decade, excessive inflow and infiltration (I/I) and a lack of effluent storage and disposal capacity resulted in overflows from the effluent storage reservoir during periods of above average rainfall. Such overflows were to Lyons Creek, then Clear Lake, which were in violation of WDRs Order No. 98-206 and Lake County Ordinances. In addition, monthly monitoring reports submitted pursuant to WDRs Order No. 98-206 indicated that the Discharger repeatedly violated effluent limitations for various water quality parameters. Cease and Desist Order (CDO) No. 94-064 was adopted by the Board on 25 February 1994 and contained a time schedule for the Discharger to expand its treatment system and obtain adequate disposal facilities. On 6 December 1996, the Board amended the CDO to provide a revised compliance schedule for completing the expansion project. This schedule calls for project completion by 1 September 2001. The CDO also includes a connection restriction, limiting new connections or increases in influent volume or loading. The Discharger's proposed Project, as reflected in this Order, is in response to CDO No. 94-064.

The Discharger submitted a Report of Waste Discharge (ROWD), dated 6 June 2000, for the proposed changes to its NWR wastewater collection, treatment and disposal system. Additional documentation, including the July 2000 *Title 22 Engineers Report - Recycled Water Project* (Title 22 Report), the October 1999 *Environmental Assessment/Finding of No Significant Impact - Initial Study/Mitigated Negative Declaration* (Initial Study), letter dated 1 March 2001, facsimile dated 4 April 2001, and numerous design reports were also provided by the Discharger to supplement the ROWD.

Through implementation of the Project, the Discharger will complete numerous upgrades to its collection system as well as the NWRWTP. For optimum flexibility, the Discharger's pump stations contain automatic control valves for gravity flow and remote operation, emergency release systems, and gravity bypass systems. Surge tanks prevent column separation and water hammer effects on the pumps and appurtenant pipelines.

In addition to raw wastewater delivered from the collection system, a receiving station located at the NWRWTP's headworks receives septage from local septage haulers. Proposed improvements to the treatment system under this Order include conversion of the existing oxidation ditches to fully-mixed lagoon cells. An additional 1.5-million gallon, fully-mixed lagoon cell and approximately 6.8 million gallons of partially mixed lagoons with floating baffles will also be constructed as part of the facility improvements. The Discharger also proposes to switch from chlorine gas to liquid sodium hypochlorite disinfection. Stabilized sludge from the treatment process is currently discharged to a dedicated disposal area on Discharger-owned land at the treatment plant site. A solids recycling system is proposed for improved biosolids management under this Order.

As in the previous WDRs Order No. 98-206, treated effluent will be discharged to an 865-acre-ft storage reservoir (NWR Reservoir) located at the NWRWTP. From the NWR Reservoir, effluent may be recycled through spray irrigation of the existing 540-acre recycled water irrigation area, or discharged to the proposed NWR Pump Station (NWPS No. 1). Upon completion of the Project, the Discharger does not intend to conduct spray irrigation of treated effluent under normal operating conditions. Use of the spray irrigation area will be minimal and typically as a contingency. During periods of recycled water application, all of the excess runoff from the irrigation area is collected in a tailwater control basin and returned to the NWR Reservoir. Recycled water irrigation shall not occur in areas where the public has the potential to come into contact with the recycled water, including direct spray or by movement of aerosols. Given the potential for public exposure at wetland cells constructed at the NWRWTP site, the net effective application area available for recycled water irrigation has been slightly reduced.

Treated effluent may also be discharged from the NWR Reservoir via buried pipeline to the upper-most cell in a series of four constructed wetland cells at the NWRWTP (hereafter collectively referred to as "NWR Wetland"). Referred to as NWRWTP/Lyons Creek Wetland Site No. 6 in the ROWD documents, the 5.04-acre NWR Wetland was initially constructed in July 2000 and operated as a pilot study/demonstration project. Intended primarily for wildlife habitat enhancement and restoration, the NWR Wetland is also anticipated to recycle and polish the effluent, manage storm water runoff, reduce peak flood flows, control erosion, and provide public benefits through the creation of settings for passive recreation and wildlife study.

Located out of the 100-year flood plain, 5.04-acre NWR Wetland was previously undeveloped and used for recycled water irrigation and cattle grazing. The sites' topography generally consists of flat riparian valleys with ephemeral stream channels. To take advantage of the existing topography and minimize the need for cut/fill, the Sloped Wetland Cell design was used for construction of the NWR Wetland. Varying degrees of shaping and grading have been incorporated into the wetland cell designs to create inter-mixed areas of deep water and shallow marsh habitat. The cells have also been designed with a blend of both open water and emergent marsh conditions to create ideal habitat for a broad variety of wetland-dependant birds and other forms of wildlife.

As described in a 4 April 2001 facsimile by the Discharger, the operational concept of the NWR Wetland includes gravity flow through the four serially interconnected wetland cells. At the outflow of the final cell, any excess water will be returned to the NWR Reservoir through a recapture pipeline. Storm water management ditches are to be constructed at the exterior toe of the wetland levees to divert runoff from the surrounding watershed into existing waterways. Any runoff or discharge of recycled water commingled with storm water from the constructed wetlands shall be contained within the Discharger's Designated Recycled Water Areas or returned to the NWR Reservoir.

From the NWR Reservoir, the NWPS No. 1 will lift recycled water 430 feet over a distance of 110,000 feet (20.8 miles) to an intermediate booster pump station (NWPS No. 2). Equipped with a 150,000-gallon regulating reservoir, NWPS No. 2 is adjacent to the Clearlake Oaks County Water District's (COCWD) recycled water pump station. As proposed, the regulating reservoir at NWPS No. 2 will be a bolted steel tank, 12 feet in diameter and 16 feet high.

To facilitate the delivery of recycled water from NWPS No. 1 to NWPS No. 2, a 16-inch pipeline (NWR Pipeline) will extend along the west and north shores of Clear Lake with a design flow rate of 4.0 mgd. Five major stream crossings, including Rodman Slough near NWRWTP and Schindler Creek near COCWD, exist along the pipeline alignment.

In October 1999, the COCWD completed construction of a pipeline (COCWD Pipeline) to connect the NWPS No. 2 to the SERWTP storage reservoir. Through this 16-inch diameter, 6,800-foot long pipeline, treated and disinfected wastewater from the 2.1 mgd-capacity COCWD pump station and the 4.0 mgd-capacity NWPS No. 2 is delivered to the SERWTP's storage reservoir for ultimate discharge into the SEG Pipeline.

As described in the Title 22 Report, the pump stations will consist of multiple vertical or horizontal split-case pumps with automated controls to restrict the flow in the event of an emergency. Pump operating conditions and control data will be transmitted to the Discharger's computer system for monitoring and automatic/manual operator control.

The Discharger's SERWTP is regulated under separate WDRs Order No. 96-166 which provides for the discharge of treated effluent from the SERWTP storage reservoir to the Southeast Geysers (SEG) Pipeline. The Discharger, in partnership with geothermal steam suppliers and users, constructed the SEG Pipeline to deliver treated and disinfected wastewater combined with supplemental lake water to a geothermal resource area known as KGRA, or the Geysers, for injection. The KGRA is located approximately 14 miles southwest of the SERWTP and since the 1970's has been actively tapped for steam production in order to generate electricity. Steam production has decreased over the years and the managed injection of water has proven to prolong the life of the steamfield in specific areas through improved steam reservoir pressures, resulting in increased power production.

Other wetland sites at the NWRWTP, SERWTP and at various locations along the length of the pipeline were initially proposed for construction as part of the Project but are no longer being considered at this time. Any future plans for recycled water use in constructed wetlands not located at the NWRWTP are to be covered under separate WDRs or Water Recycling Requirements.

As described in an August 1998 report by CH2M Hill entitled *Clear Lake Basin 2000 Preliminary Design Report - Constructed Wetlands*, pollutant removal efficiencies of the wetlands were primarily calculated as a function of influent concentrations and hydraulic loading, with reduction modeled as a first-order, area-dependent process. Given a recycled water

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effluent concentration to the wetlands of 40 mg/L BOD, the predicted effluent quality from the wetland would be 6 mg/L BOD. This estimate is based upon an average annual loading to the wetlands of approximately 8,400 gpd/acre of wetland.

Wastewater effluent from the NWR treatment system is referred to as effluent. Effluent discharged from the NWR Reservoir to the irrigation area, NWR Wetland, NWR Pipeline and/or COCWD Pipeline is referred to as recycled water.

The Discharger has the responsibility for the construction, operation, maintenance and monitoring of the two recycled water pump stations (NWPS No. 1 and NWPS No. 2), NWR Pipeline and the NWR Wetland. This Order establishes requirements for the Discharger's NWR collection system, NWRWTP, NWR Reservoir, NWR Wetlands, NWR Pipeline, and NWPS Nos. 1 and 2. Regulation of the COCWD Pipeline and any other constructed wetlands will be regulated under separate requirements.

The NWRWTP, NWR Reservoir, 540-acre on-site irrigation disposal area, and NWR Wetland are in Sections 25, 26, 35 and 36, T15N, R10W, MDB&M (Assessor's Parcel No. 003-046-30). The NWR Wetland regulated under this Order is owned by the Discharger and is shown on Attachment A, which is attached hereto and part of the Order by reference.

The Basin Plan encourages reclamation. Constructed wetlands (providing recreation and wildlife) and the injection of blended effluent and lake water for steam/electric production are encouraged reclamation projects.

The Discharger submitted an engineering report, dated July 2000, in accordance with the California Code of Regulations (CCR) Title 22, Division 4, Chapter 3, Article 7, Section 60323, Water Reclamation Criteria.

The Board consulted with the California Department of Health Services, Lake County Health Department, California Department of Fish and Game, and Mosquito Abatement District and considered their recommendations regarding public health aspects for use of recycled water.

In August 1997, the Discharger's Board of Directors certified the *Northwest Regional Wastewater Treatment and Disposal Facility Improvements EIR* (EIR), which examined proposed improvements for collection system, treatment systems, recycled effluent storage, and reuse. Subsequent to approval of the abovementioned EIR, the Discharger identified additional wetland sites for implementation along the State Route 20 corridor that were not previously identified in the analysis. Therefore, on 2 October 1999 the U.S. Environmental Protection Agency adopted an October 1999 *Clearlake Basin 2000 - Phase II Environmental Assessment/Finding of No Significant Impact and Initial Study/Mitigated Negative Declaration* in accordance with the National Environmental Policy Act (NEPA). The Discharger's Board of Directors approved an Initial Study/Mitigated Negative Declaration on 7 February 2000 in accordance with the California Environmental Quality Act (CEQA) (Public Resources Code

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Section 21000, et. seq.) and the State CEQA Guidelines. As stated in the Mitigated Negative Declaration, the project would not have a significant effect on the environment, because mitigation measures have been incorporated into the proposed project.

Lyons Creek drains the entire NWRWTP and recycled water irrigation areas. Drainage north of ridges north of the site flows to Robinson Creek and then to Rodman Slough north of and tributary to Clear Lake. Lyons Creek drainage is confined by a shallow runoff retention basin at the point it leaves the site.

The beneficial uses of Lyons Creek, Clear Lake and Cache Creek are domestic, municipal, industrial and agricultural supply; recreation; aesthetic enjoyment; navigation; ground water recharge; fresh water replenishment; hydroelectric power generation; and preservation and enhancement of fish, wildlife, and other aquatic resources.

The beneficial uses of underlying groundwater are domestic, municipal, industrial, and agricultural supply.

Surface water drainage is to Lyons Creek, which is tributary to Clear Lake, thence to Cache Creek.

SKC: 11 May 2001